Jack Leckert

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PROFILE

Software Engineer and UC Berkeley graduate, specializing in building and deploying production-grade software for autonomy applications.

RESEARCH INTERESTS	TECHNICAL SKILLS	PROFESSIONAL SKILLS
ML and applied AI	Python, C/C++, Go, TypeScript	French – native language
Model and learning based controls	Linux / CLI	English – fluent
Localization	Github / CI	German – fluent
	AWS, Kubernetes, Jenkins	

PROFESSIONAL EXPERIENCE

Member of Technical Staff, Product OCTOBER 2026 – PRESENT
Salient, San Francisco, CA, USA
Software engineer building AI voice agents for consumer lending.

Software engineer JULY 2024 – SEPTEMBER 2026
Swift Navigation, San Francisco, CA, USA
Software engineer as part of the IoT team.
- Proposed, led and built an automated infrastructure monitoring platform to optimize the quality and availability of Swift's base stations network. This effort led to the acquisition of 2 new customers.
- Fine-tuned Starling positioning engine parameters using a genetic algorithm, leading to a 20% increase in 2D position accuracy of Swift's XGPS DashPro.
- Designed and built a suite of customer-facing REST APIs for real-time service indicators (geofencing, coverage).
- Developed various protocol converters for data processing and visualization.

Graduate student instructor AUGUST 2023 – JUNE 2024
UC Berkeley, CA, USA
Teaching assistant for the undergraduate class Experimentation and Measurements.
- Preparing and supervising weekly labs of 20+ students, along 2 other GSIs.

Software engineer FEBRUARY 2023 – JUNE 2023
Skybase, Christchurch, New Zealand
Flight software engineer with the mission to develop a vision system for an autonomous fire retardant airplane.
- Developed automated object detection functions for a remotely piloted flight control system.

Technology analyst intern MAY 2022 – OCTOBER 2022
Starburst Aerospace, Munich, Germany
Technology analyst intern as part of the Starburst Startup Accelerator in Munich.
- Conducted one major consulting project about hot and harsh environment electronics for a European Tier-2 Aerospace supplier.
- Assessed the technology viability of Aerospace startups and fit for the Starburst community (+300 startups).

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EDUCATION

Master of Engineering, Robotics
University of California, Berkeley, USA
Focus on controls and machine learning. GPA: 3.96/4.00.
Lean LaunchPad.

Engineering degree, Robotics & Mechanics
École Centrale de Lyon, France
Computer science, machine learning, non-linear automatic control, numerical analysis, mathematics, solid and fluid mechanics, physics, material science, signal processing, electronics. GPA: 3.78/4.00.
President of the Junior Enterprise.

Preparatory classes for the Grandes Ecoles
Lycée du Parc, Lyon, France
Two years of intensive preparatory classes for the French engineering schools nationwide scientific competitive examination (MPSI and PSI*).
Mathematics, physics, mechanics, automatic control, computer science, chemistry.

High School
Lycée Français Victor Hugo, Frankfurt, Germany
Scientific Baccalauréat (high school diploma), highest honours.
One trimester of the 10th grade spent at Winnipeg, Canada.

RESEARCH AND ENGINEERING EXPERIENCE

UAV research project SEPTEMBER 2023 – JUNE 2024
UC Berkeley, Hybrid Robotics lab
Research project focusing on developing a controller for a flapping wing UAV using model-based control and reinforcement learning. Simulation software used is Mujoco.
- Drone design developed on Solidworks and converted to XML.
- Computation of fluid forces in C++ using aero strip theory (Wagner's model).
- MPC and RL controller developed in Python for the simulation.

ESA AutoICE challenge FEBRUARY 2023 – MAY 2023
Deep learning challenge in partnership with DTU to automate sea ice mapping of the Artic using Sentinel-1 SAR imageries.
- Selection of three main parameters to qualify: ice concentration, stage of development and floe size.
- Trained a U-net (contracting and expanding CNN) using parallel cloud computing from Azure.
- By fine-tuning hyperparameters, I reached F1 and R2 scores of about 80%.

Fluid mechanics research project SEPTEMBER 2021 – MAY 2022
École Centrale de Lyon, Fluid Mechanics and Acoustics Laboratory (LMFA)
Research study focusing on reducing the drag force on a marathon runner by placing partners in a specific formation around him.
- Designed and machined a test bench using CAD and Fusion 360.
- Tested experimentally and analyzed the effects on the drag force when varying parameters (distance and angles) between the runners.
- Designed a formation by iteration allowing to save up to 2min 23s on a full marathon, or 27s less than the actual world record from E. Kipchoge during the INEOS 1:59 Challenge in 2019.

A research paper has been published in the Proceedings of the Royal Society A.

Aerospace engineering project SEPTEMBER 2020 – JUNE 2021
École Centrale de Lyon, Fluid Mechanics and Acoustics Laboratory (LMFA)
Building of a supersonic rocket (the Centrale Transonic Rocket) with a team of 5 students as part of the national competition C'Space organised by Planète Sciences.
- Designed and built an electronic acquisition chain to save pressure data using C++.
- CAD designed and 3D-printed a rocket cone.